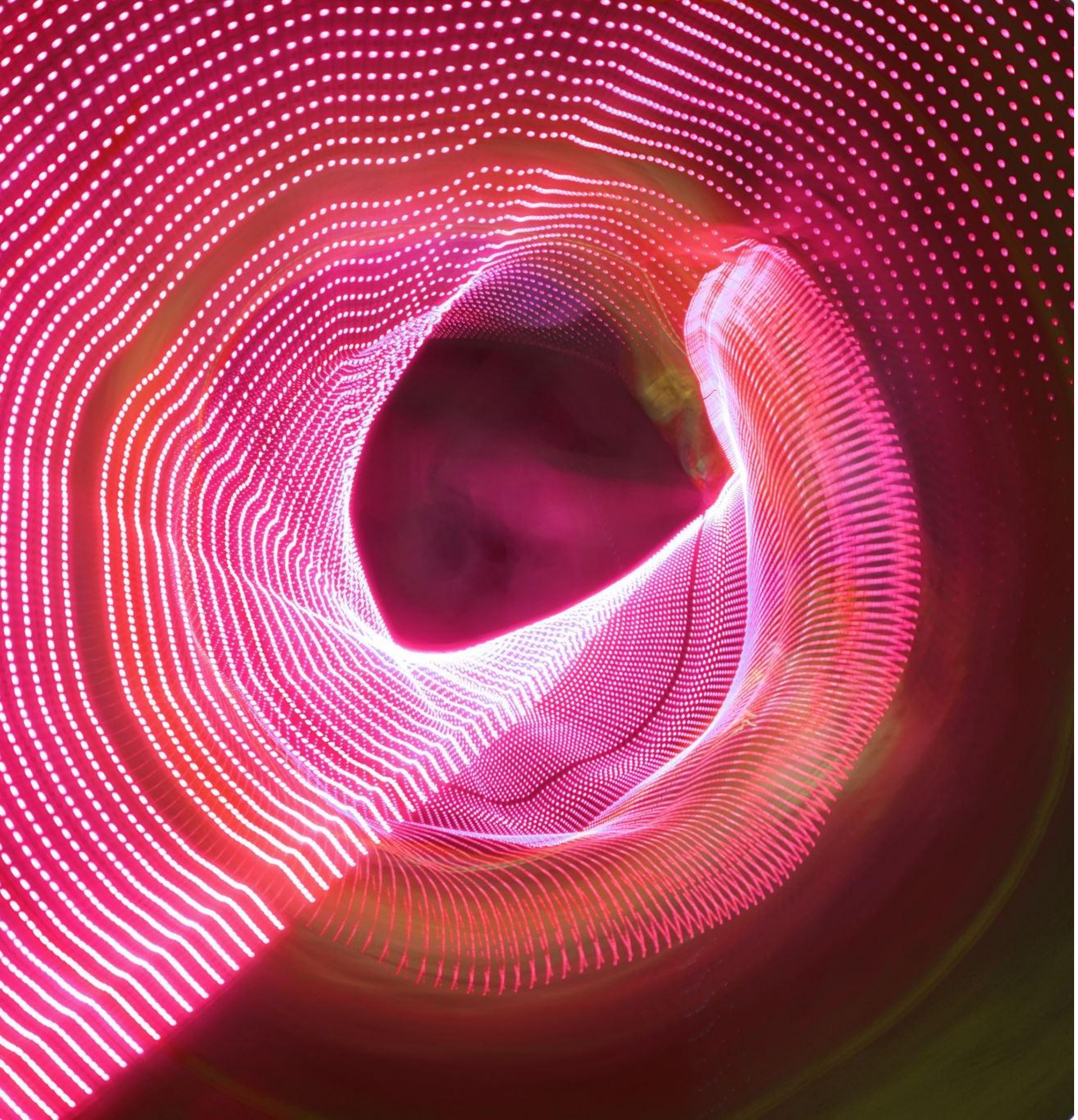

Haemorrhoids & Anal Fistula

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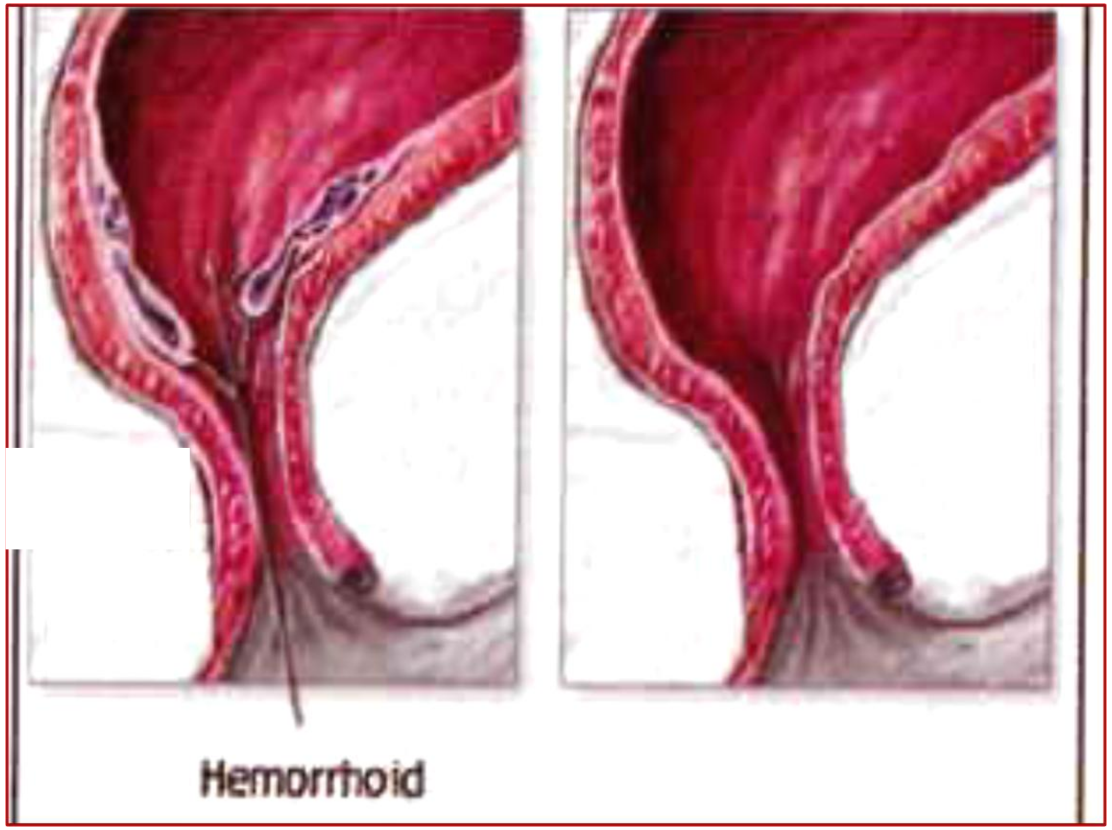
Key facts

Broad term, often incorrectly used to refer any perianal excess tissue.

True haemorrhoids are excessive amounts of the normal endoanal cushions that comprise anorectal mucosa, submucosal tissue, and submucosal blood vessels (small arterioles and veins).

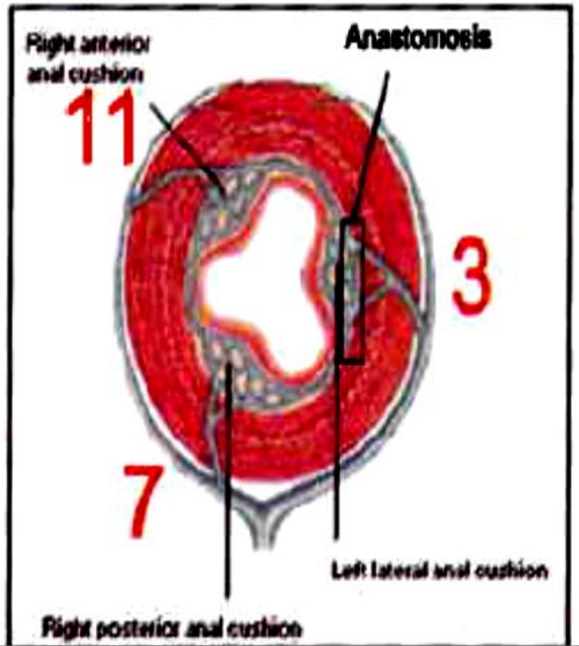
Commonest age of onset is in young adulthood.

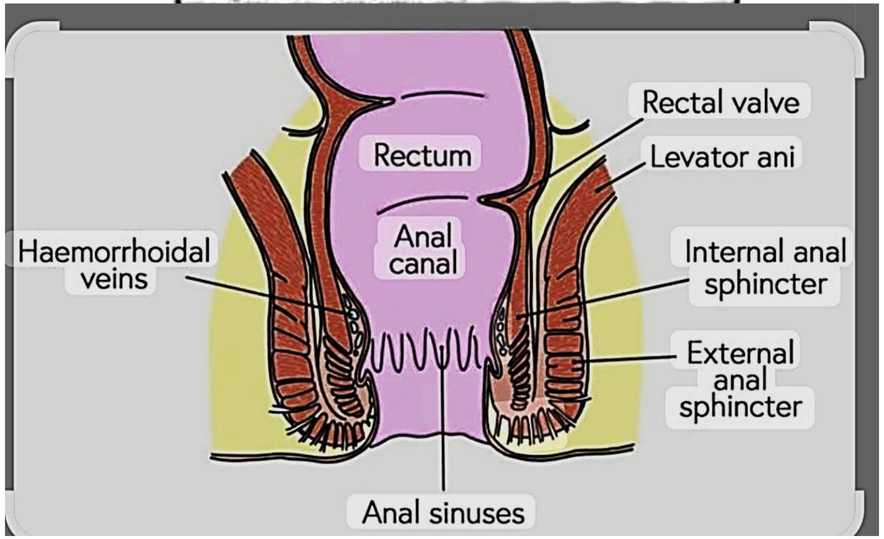
Associated with constipation, chronic straining, obesity, and previous childbirth.



Anatomy

Normally, the terminal branches of superior rectal vessels (portal) form vascular plexus with middle and inferior rectal vessels (systemic) beneath the epithelium lining of anal canal called (anal cushions) which are arranged at the 3, 7, 11 O'clock position around the circumference of the anal canal, so any congestion, enlargement and prolapse of these A – V cushions constitute piles.



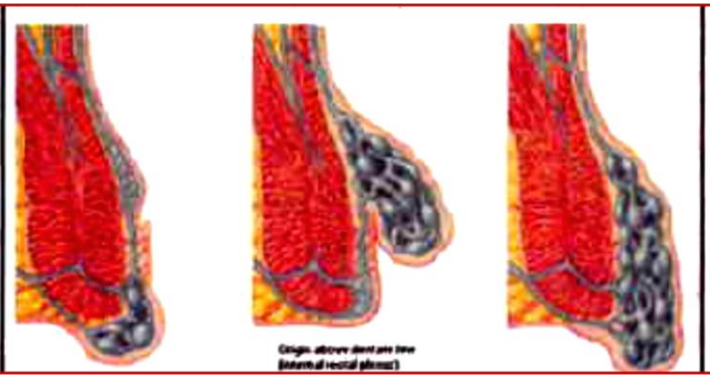


Types

- May become ulcerated and inflamed if recurrently prolapsing.
- If confined to the tissue of the upper anal canal, they are referred to as ‘internal’.
- If extend to the tissues of the lower anal canal, they are referred to as ‘external’.

Types

A- Internal piles = above dentate line and covered by mucosa.
B- External piles = below dentate lined and covered by skin.
C- Intero-external piles (the most common).



Clinical features

- ***Features of irritation:*** Pruritus ani, mucus discharge, and perianal discomfort.

- ***Features of damage to mucosal lining:*** Recurrent post-defecatory bleeding – bright red, not mixed with stools, on paper or splashing in the toilet pan.

- ***Features of prolapse:*** Intermittent lump appearing at anal margin, usually after defecation, may spontaneously reduce or require manual reduction.

Clinical Degrees of Piles According to Prolapse

First degree

- The patient has **only bleeding** but no prolapse of piles, diagnosed only by **proctoscope** as it is not felt by DRE. It is not reaching level of dentate line.

Second degree

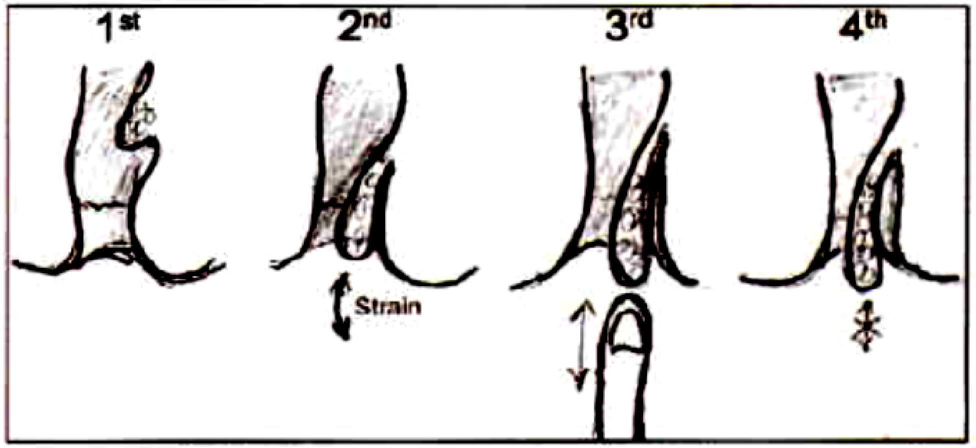
- The piles prolapse only during defecation, but they are spontaneously reduced at the end of act.

Third degree

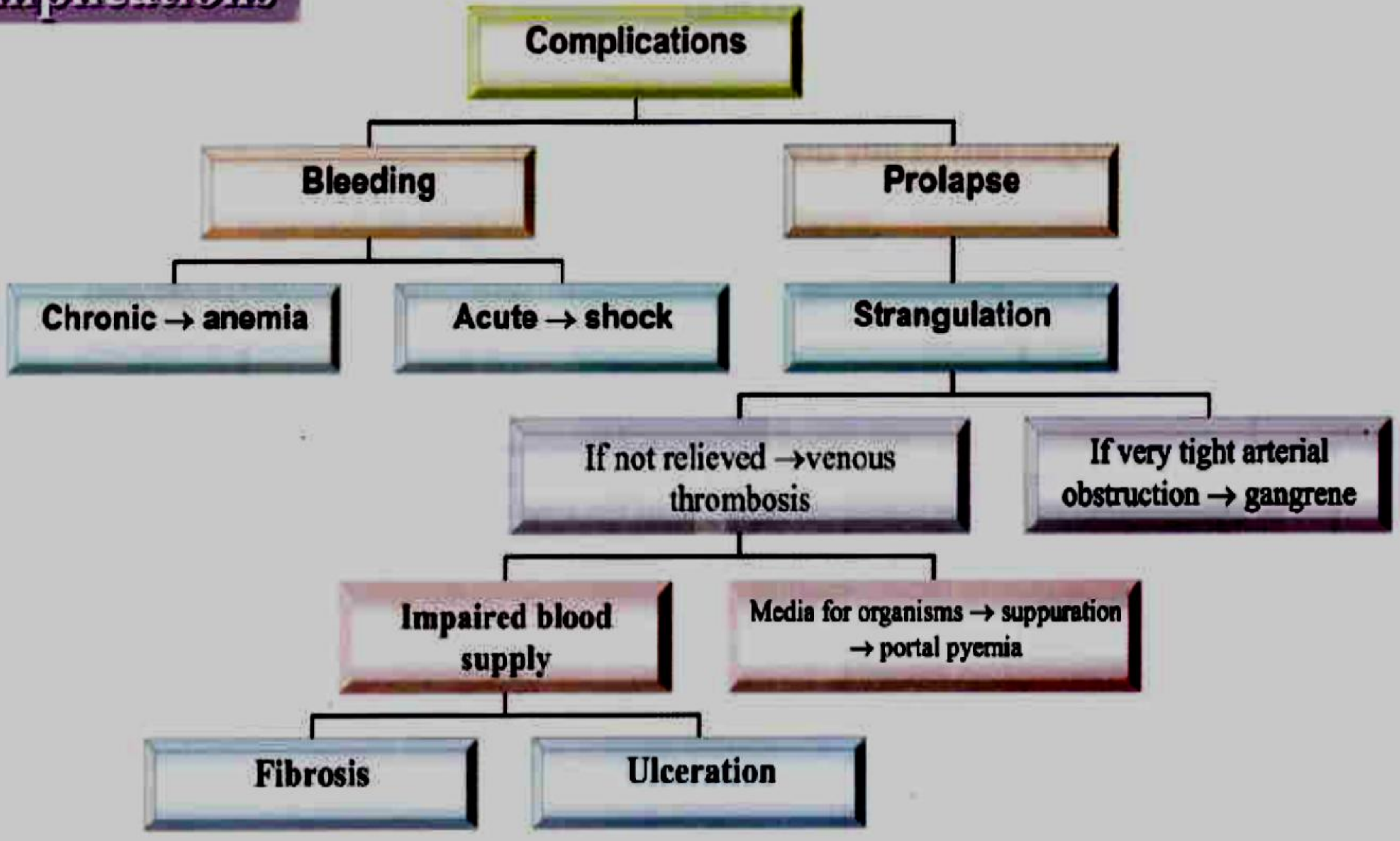
- There is prolapse of piles during defecation and patient has to reduce it manually.

Fourth degree

- There is permanent prolapse of piles.



Complications



Diagnosis and investigations

- Diagnosis is usually by **rigid sigmoidoscopy** and **proctoscopy**.

- Flexible sigmoidoscopy or colonoscopy may be appropriate if there is concern about the cause of symptoms; *remember* – haemorrhoids rarely start over the age of 55y and it is often best to assume another cause until proven otherwise in these cases.

Treatment

- **Uncomplicated**
  - 1ry
    - ***1st & 2nd degrees:***
      - Conservative
      - Injection sclerotherapy
      - Band ligation
    - ***3rd & 4th degrees:***
      - Hemorrhoidectomy
  - 2ry
    - TTT of the cause
- **Complicated**
  - Strangulated
  - Thrombosed
  - Infected
  - Bleeding

Treatment

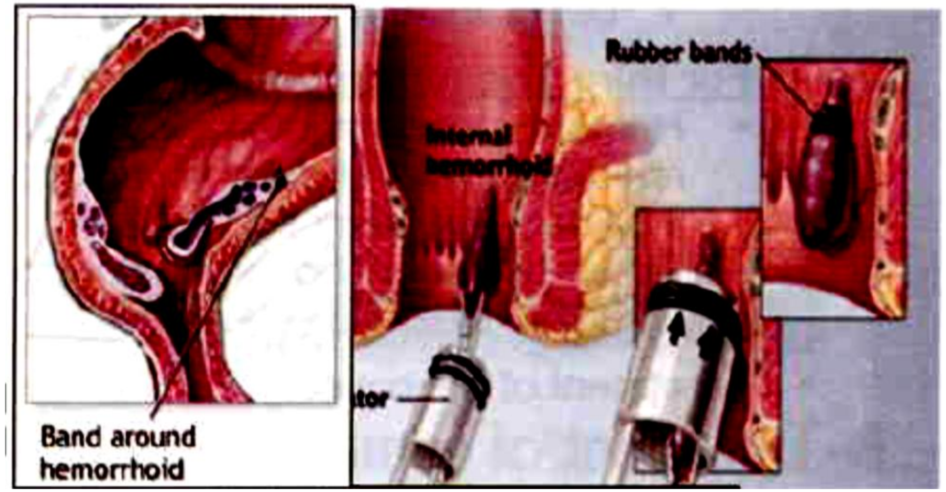
Medical treatment : Avoidance of constipation and straining; bulking or softener laxatives.

Surgical treatment

1. Banding or excessive tissue. ***Best for prolapse symptoms*** (not possible for external components due to excellent nerve supply of lower anal canal).

2. Dilute phenol injections (5% in almond oil). ***Best for bleeding symptoms***.

Treatment
(Continued)



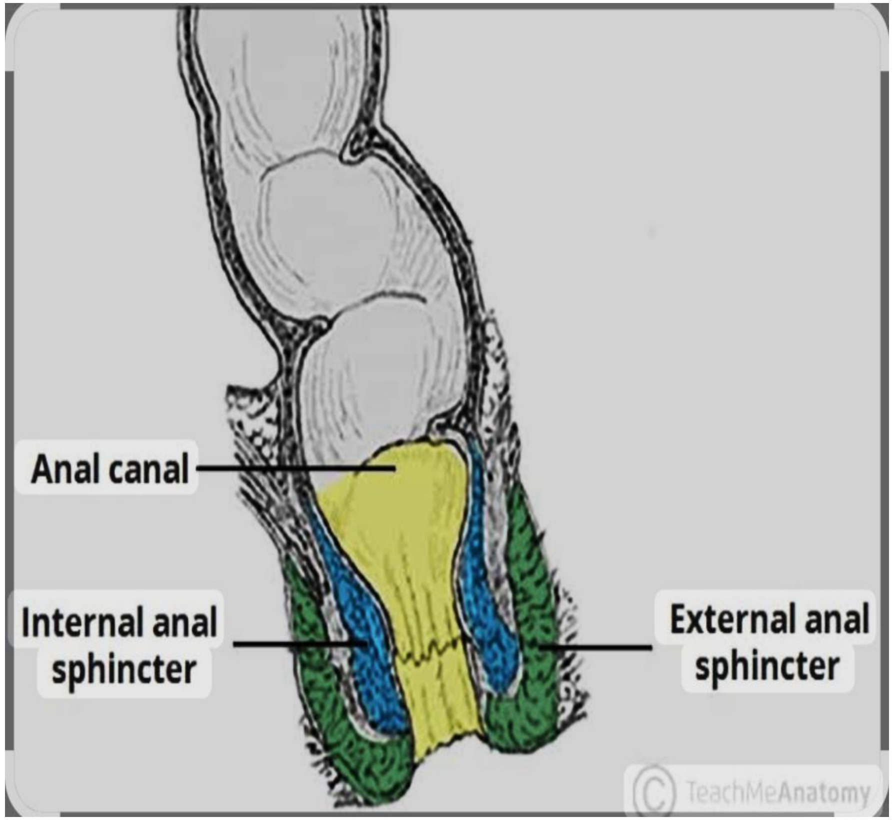
3. Haemorrhoidal devascularization procedures (e.g. arterial ligation 'HALO' either Doppler ultrasound-guided or blind). ***For failed topical treatments and surgery not indicated/desired***.

4. Haemorrhoidectomy ***for large external haemorrhoids or haemorrhoids failing to respond to conservative treatment.*** Associated with a small risk of impaired continence and anal stenosis.

5. Stapled anopexy (also called PPH). Sometimes used for circumferential prolapsing haemorrhoids.

HALO; Harmorrhoidal Artery Ligation Operation

Fistula-in-ano

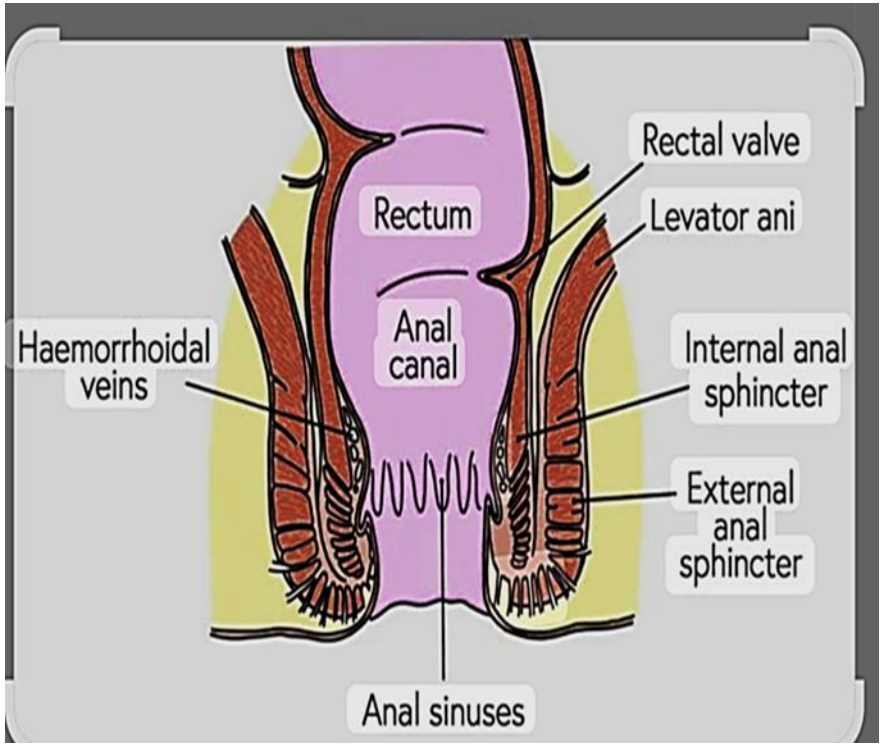


Key facts

- A fistula is an abnormal connection of two epithelial surfaces and the two surfaces joined in fistula in ano are the anorectal lining and the perineal or vaginal skin.
- Very common, especially in otherwise fit young adults.
- May occur in the presence of Crohn's disease; minor association with obesity and diabetes mellitus, very rarely due to trauma or ulceration of anorectal tumours.

Key Aspects of Anal Sphincters:

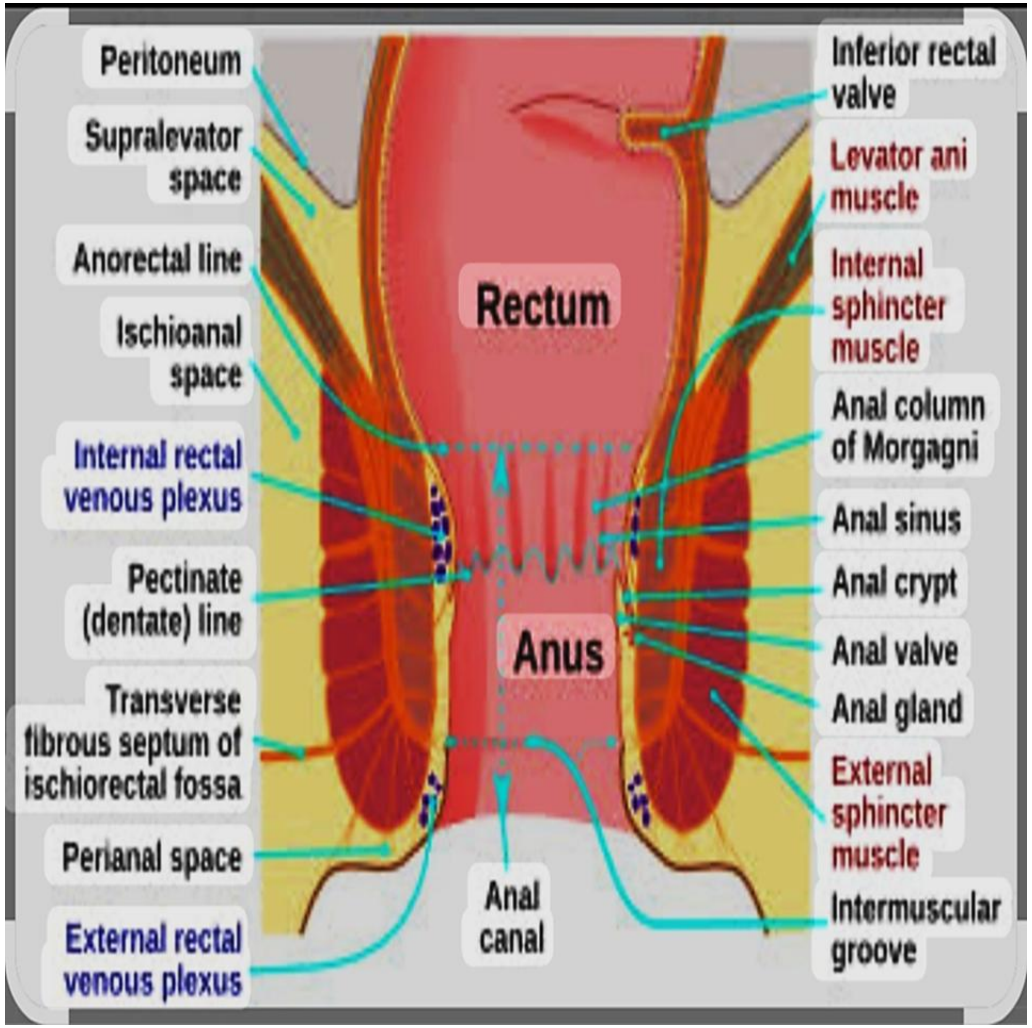
- **Internal Anal Sphincter (IAS):** A specialized thickening of the inner smooth muscle of the large intestine. It is under involuntary (autonomic) control and provides 50-85% of the resting anal pressure, keeping the anus closed.
- **External Anal Sphincter (EAS):** A striated muscle wrapping around the IAS that allows voluntary control to delay defecation. It is innervated by the pudendal nerve (S2–S4).
- **Defecation Mechanism:** When the rectum fills, the IAS relaxes (rectoanal inhibitory reflex), causing the urge to defecate. The EAS is then used to voluntarily maintain continence until a toilet is reached.
- **Location:** Situated at the end of the anal canal, surrounded by the pelvic floor muscles.



- **Clinical Relevance:** Damage, weakness, or surgery on these muscles can lead to fecal incontinence. Common conditions include hemorrhoids or anal fissures, which may be treated by addressing sphincter tone.

Anal Sphincter Anatomy Comparison

Feature	Internal Sphincter (IAS)	External Sphincter (EAS)
Muscle Type	Smooth (Involuntary)	Skeletal (Voluntary)
Primary Function	Resting tone/Continence	Voluntary control/Defecation
Control	Autonomic nervous system	Pudendal nerve

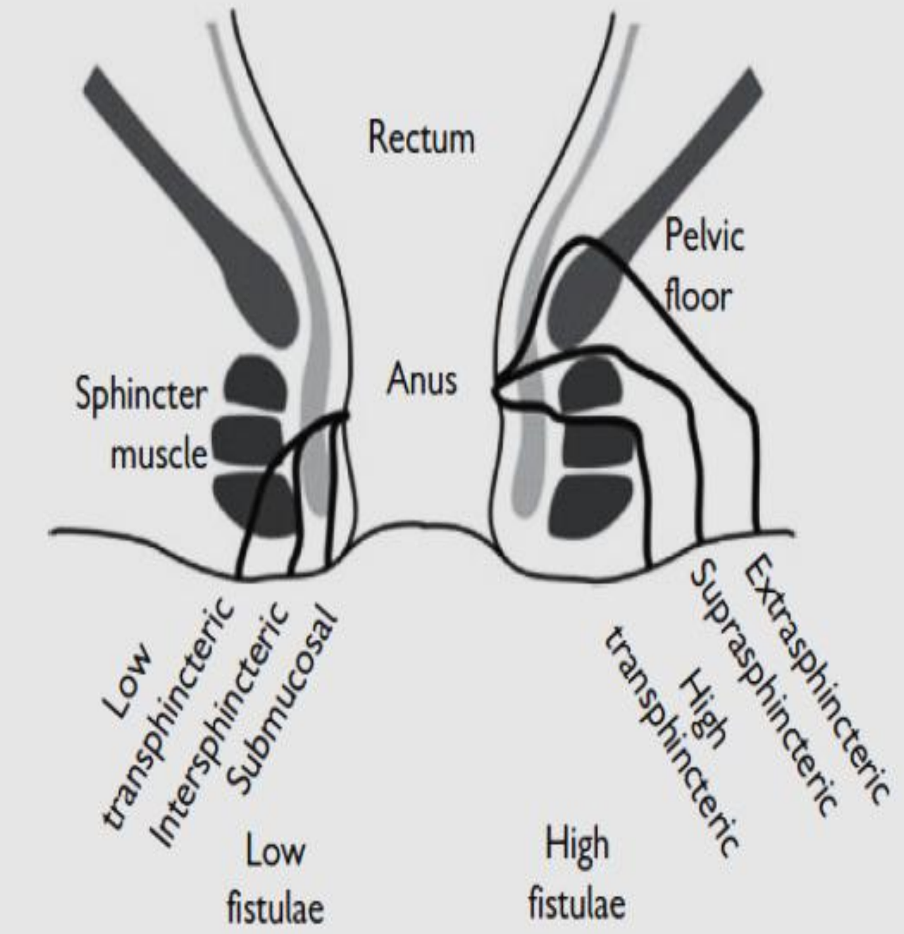


Pathological features

Commonest cause is **sepsis** arising in an anal gland that forces its way out through the anal tissues to appear in the perianal or in women, vaginal skin (cryptoglandular theory of fistula in ano).

Often presents initially as an acute perianal abscess.

The tissues through which the track pushes determines the classification of fistulas.



Classification of fistula-in-ano.

New classification (Park's classification)

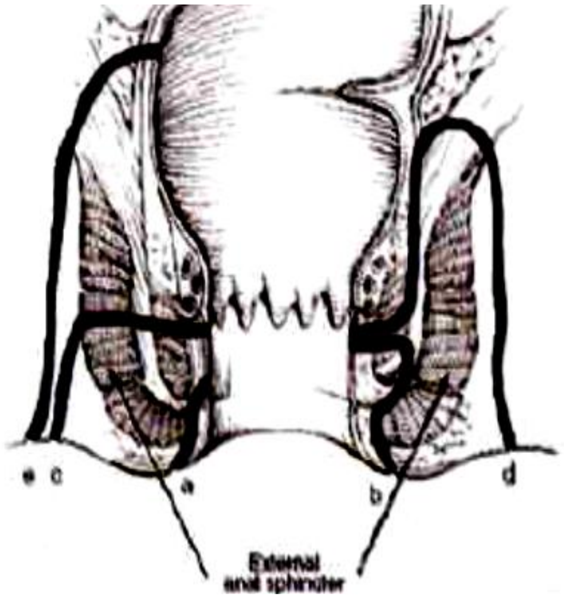
- **According to the course of fistulous track:**

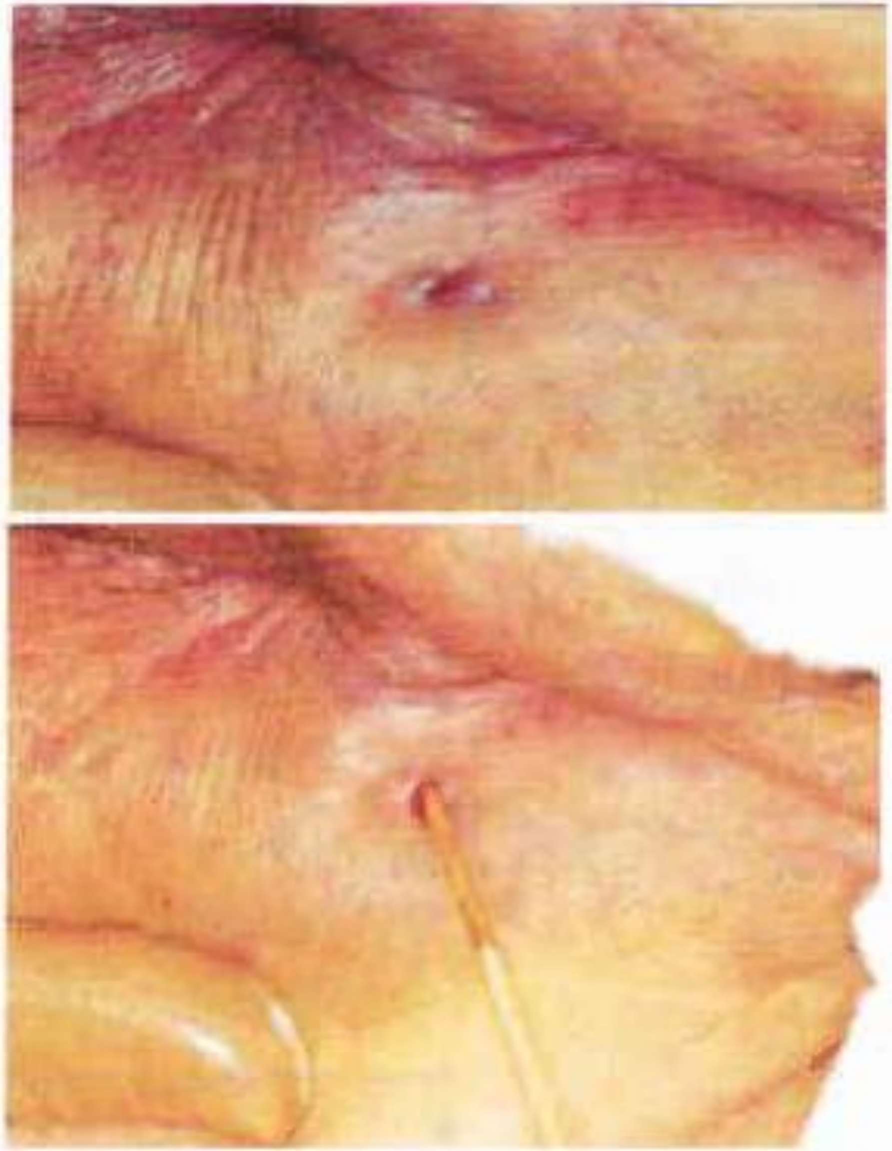
A- **An intersphincteric fistula** tracks (most common 70%) between the internal anal sphincter and the external anal sphincter in the intersphincteric space.

B- **A transsphincteric fistula** tracks from the intersphincteric space through the external anal sphincter.

C- **A suprasphincteric fistula** leaves the intersphincteric space over the top of the puborectalis and penetrates the levator muscle before tracking down to the skin.

D- **An extrasphincteric fistula** tracks outside of the external anal sphincter and penetrates the levator muscle into the rectum.





Clinical features

- ***Acute perianal abscess.*** Rapid onset of severe perianal or perineal pain.
  - Swelling and erythema of the perianal skin with fever and tachycardia.
- ***Recurrent perianal sepsis.*** Recurrent intermittent sepsis typified by gradual build-up of 'pressure' sensation and swelling in the perianal skin and eventual discharge of bloodstained purulent fluid.
- ***Chronic perianal discharge.*** Persistent low grade sepsis of the track with chronic discharge of seropurulent fluid via a punctum that is usually clearly identified by the patient.

Diagnosis and investigations

Diagnosis and investigation should aim to confirm the presence of a fistula and identify the course of the track to determine the type of fistula:

- Examination of the perineum and rectal examination may reveal a palpable fibrous track.
- **Probing** of any external opening to aid identification of the course of the track.
- **Endoanal ultrasound** (sometimes with hydrogen peroxide injected into the track) identifies the course of the track.
- **MRI scanning** is probably the most sensitive method of determining the course of the track and identifying any occult perianal or pelvic sepsis.
- **Flexible sigmoidoscopy** if associated colorectal disease, e.g. Crohn’s disease, is suspected.

Treatment

Aim

- Eradication of sepsis with preservation of anorectal function.

Options

1- **Low fistula:**

- Lay open (fistulotomy☑) or fistulectomy.

2- **High fistula:**

- Lay open for the lower part
- The upper part

1. Seton wire (staged fistulotomy) or
2. Fibrin glue to avoid the 2^{nd} stage

If failed →

Proximal colostomy (to divert the stools away from the fistula)

Exception☑

High inter-sphincteric fistula → lay open fistulotomy for both parts

Surgical treatment

Principles of surgical treatment are as follows:

- Drainage of any acute sepsis if present.
- Prevention of recurrent sepsis. Usually by insertion of a loose seton suture, e.g. silastic sling.
- **Low fistula in ano:** Lay open track, remove all chronic granulation tissue, and allow to heal spontaneously (fistulotomy); little risk of impairment of continence due to minimal division of sphincter tissues.
- **High fistula in ano:**
- Remove fistula track and close the internal opening (core fistulectomy and endorectal flap advancement).
- Slowly divide the sphincter tissue between the fistula and the perianal skin (cutting seton); low risk of incontinence.
- Fill the fistula with fibrin glue.


Thank
You